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Towards improving synthetic reporters for phenotypic mapping of glioblastoma states by LSD+

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Cell types and cell states are governed by complex gene regulatory programs encoded in cis-regulatory DNA. In glioblastoma (GBM), diverse cellular states contribute substantially to intratumoral heterogeneity and therapeutic resistance, yet their functional characterization remains challenging. To address this, we developed LSD+ (Logical Design of Synthetic cis-regulatory DNA +), a computational framework for designing synthetic locus control regions (sLCRs) that translate transcriptional signatures into compact and functional regulatory DNA constructs.

To validate the framework, we tested whether LSD+ could recover known natural locus control regions using phenotype-specific transcription factor programs. Using transcription factors derived from yolk sac erythroid progenitors (YEPs), LSD+ identified significant enrichments of known regulatory elements within the alpha- and beta-globin loci. Conversely, cone cell-derived transcription factors failed to enrich these erythroid regulatory regions but successfully identified regulatory elements within the opsin locus. Systematic on-target and off-target analyses across alpha-globin, beta-globin, and opsin loci demonstrated that phenotype matched transcription factor programs consistently recover known natural locus control regions significantly above random expectation, whereas mismatched programs do not. These results support LSD+ as a generalizable framework for discovering functional cis-regulatory architectures from phenotype-specific transcriptional programs, particularly in contexts where natural regulatory elements remain unknown.



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